Worksheet 3.2 — Derivatives of Exponential Functions

Grade 12 Calculus and Vectors (MCV4U) · Unit 3: Derivatives of Transcendental Functions

The Big Ideas

$\frac{d}{dx}e^x = e^x$ (its own derivative); $\frac{d}{dx}b^x = b^x \ln b$; with the chain rule $\frac{d}{dx}e^g = e^g g'$.

- $\frac{d}{dx}e^x = e^x$.
- $\frac{d}{dx}b^x = b^x \ln b$.
- $\frac{d}{dx}e^{g(x)} = e^{g(x)}g'(x)$.

The Big Picture

The special base e

The exponential e^x is its own derivative: $\frac{d}{dx}e^x = e^x$, which is why e is calculus's natural base.

Other bases

For a general base, $\frac{d}{dx}b^x = b^x \ln b$ the extra factor $\ln b$ appears.

With the chain rule

$\frac{d}{dx}e^{g(x)} = e^{g(x)}\,g'(x)$, multiplying by the derivative of the exponent.

Worked Examples

Example 1 — Base e

Differentiate $f(x) = e^x$.

Solution. **Step 1 (recall the special property of e^x):** $\frac{d}{dx}e^x = e^x$ — it is its own derivative.

Conclusion: $f'(x) = e^x$ — the slope equals the height at every point:

Example 2 — Chain rule

Differentiate $f(x) = e^{3x}$.

Solution. **Step 1 (identify outer and inner):** outer is $e^{(\cdot)}$, inner is $g(x) = 3x$ (so $g'(x) = 3$).
Step 2 (differentiate the outer, using $\frac{d}{dx}e^u = e^u$): e^{3x}.
Step 3 (multiply by the inner derivative): $e^{3x} \cdot 3$.
Conclusion: $f'(x) = 3e^{3x}$.

Example 3 — Other base

Differentiate $f(x) = 2^x$.

Solution. **Step 1 (recall the rule for a general base):** $\dfrac{d}{dx}b^x = b^x \ln b$; here $b = 2$.
Conclusion: $f'(x) = 2^x \ln 2$.

Example 4 — Product rule

Differentiate $f(x) = xe^x$.

Solution. **Step 1 (identify the two factors):** $u = x$ (so $u' = 1$) and $v = e^x$ (so $v' = e^x$).
Step 2 (apply the product rule $u'v + uv'$): $(1)e^x + xe^x$.
Step 3 (factor out e^x): $e^x(1 + x)$.
Conclusion: $f'(x) = e^x(1 + x)$.

Example 5 — Chain (square inside)

Differentiate $f(x) = e^{x^2}$.

Solution. **Step 1 (identify outer and inner):** outer is $e^{(\cdot)}$, inner is $g(x) = x^2$ (so $g'(x) = 2x$).
Step 2 (differentiate the outer, multiply by the inner derivative): $e^{x^2} \cdot 2x$.
Conclusion: $f'(x) = 2x\, e^{x^2}$.

Example 6 — Chain rule

Differentiate $f(x) = e^{5x}$.

Solution. **Step 1 (identify outer and inner):** inner is $g(x) = 5x$ (so $g'(x) = 5$).
Step 2 (differentiate the outer, multiply by the inner derivative): $e^{5x} \cdot 5$.
Conclusion: $f'(x) = 5e^{5x}$.

Example 7 — Other base

Differentiate $f(x) = 3^x$.

Solution. **Step 1 (apply the general-base rule $\frac{d}{dx}b^x = b^x \ln b$):** here $b = 3$.
Conclusion: $f'(x) = 3^x \ln 3$.

Example 8 — Negative exponent

Differentiate $f(x) = e^{-x}$.

Solution. **Step 1 (identify outer and inner):** outer is $e^{(\cdot)}$, inner is $g(x) = -x$ (so $g'(x) = -1$).
Step 2 (differentiate the outer, multiply by the inner derivative): $e^{-x} \cdot (-1)$.
Conclusion: $f'(x) = -e^{-x}$.

Example 9 — Product rule

Differentiate $f(x) = x^2 e^x$.

Solution. **Step 1 (identify the two factors):** $u = x^2$ (so $u' = 2x$) and $v = e^x$ (so $v' = e^x$).
Step 2 (apply the product rule): $2x\,e^x + x^2 e^x$.
Step 3 (factor out e^x): $e^x(x^2 + 2x)$.
Conclusion: $f'(x) = e^x(x^2 + 2x)$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Differentiate $f(x) = e^{5x}$.

Question 2

Differentiate $f(x) = 3^x$.

Question 3

Differentiate $f(x) = x^2 e^x$.

Question 4

Differentiate $f(x) = e^{-x}$.

Question 5

Differentiate $f(x) = e^{x^3+1}$.

Question 6

Differentiate $f(x) = 4e^x$.

Question 7

Differentiate $f(x) = e^{2x}$.

Question 8

Differentiate $f(x) = 5^x$.

Question 9

Differentiate $f(x) = xe^{2x}$ (product + chain).

Question 10

Differentiate $f(x) = e^{-3x}$.

Question 11

Slope of $y = e^x$ at $x = 0$?

Question 12

Differentiate $f(x) = e^x + x^2$.

Question 13 — Challenge

Differentiate $f(x) = \dfrac{e^x}{x}$ (quotient rule).

Question Recap & Answer Key

Question Recap

1. Differentiate $f(x) = e^{5x}$.
2. Differentiate $f(x) = 3^x$.
3. Differentiate $f(x) = x^2 e^x$.
4. Differentiate $f(x) = e^{-x}$.
5. Differentiate $f(x) = e^{x^3+1}$.
6. Differentiate $f(x) = 4e^x$.
7. Differentiate $f(x) = e^{2x}$.
8. Differentiate $f(x) = 5^x$.
9. Differentiate $f(x) = xe^{2x}$ (product + chain).
10. Differentiate $f(x) = e^{-3x}$.
11. Slope of $y = e^x$ at $x = 0$?
12. Differentiate $f(x) = e^x + x^2$.
13. Differentiate $f(x) = \dfrac{e^x}{x}$ (quotient rule).

✓ Answer Key

1. $5e^{5x}$
2. $3^x \ln 3$
3. $e^x(x^2 + 2x)$
4. $-e^{-x}$
5. $3x^2 e^{x^3+1}$
6. $4e^x$
7. $2e^{2x}$
8. $5^x \ln 5$
9. $e^{2x}(1 + 2x)$
10. $-3e^{-3x}$
11. 1
12. $e^x + 2x$
13. $\dfrac{e^x(x-1)}{x^2}$